Ordinary Differential Equation

- **Ordinary Differential Equations (ODE)** relate a function $y(t)$ to its derivatives.
- An **Autonomous** ODE has no explicit dependence on t.
- The order of the highest derivative in an ODE is the **Order**.
- A first order **Initial Value Problem (IVP)** is an ODE with an **Initial Condition (IC)** $y(t_0) = y_0$.
- A function satisfying the IC and the ODE for $a < t_0 < b$ solves the IVP on (a, b).

Example: Solution of a First-Order IVP

Solve the IVP $y' = 2y$, $y(0) = 3$.

The solution of the IVP is $\boxed{y(t) = 3e^{2t}}$

Check: $y'(t) = 6e^{2t} = 2y(t)$ and $y(0) = 3$.

Second Order to First Order

1. Start with a second-order IVP

$$y'' = g(t, y, y'), \qquad y(t_0) = a_1, \qquad y'(t_0) = a_2$$

2. Define $y_1 = y$ and $y_2 = y'$. Then

$$\begin{aligned} y_1' &= y_2 & y_1(t_0) &= a_1 \\ y_2' &= g(t, y_1, y_2) & y_2(t_0) &= a_2 \end{aligned}$$

3. With $\boldsymbol{y} = (y_1, y_2)$ and $\boldsymbol{y}_0 = (a_1, a_2)$, write

$$\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y}), \qquad \boldsymbol{y}(t_0) = \boldsymbol{y}_0, \qquad \boldsymbol{f}(t, \boldsymbol{y}) = \begin{pmatrix} y_2 \\ g(t, y_1, y_2) \end{pmatrix}$$

Mass and Spring

Example: Second Order to First Order

- A mass $m > 0$, damping coefficient $c \geq 0$, spring stiffness $k > 0$, and applied force $g(t)$ give

$$my'' + cy' + ky = g(t)$$

- Displacement $y_1 = y$ and velocity $y_2 = y'$ give

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}'}_{\boldsymbol{y}'} = \underbrace{\begin{pmatrix} y_2 \\ \big(g(t) - cy_2 - ky_1\big)/m \end{pmatrix}}_{\boldsymbol{f}(t,\boldsymbol{y})} \iff \boxed{\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y})}$$

- Initial displacement and velocity supply the two initial values.

Existence and Uniqueness

First-order IVP

If $\boldsymbol{f}$ and its Jacobian $D_{\boldsymbol{y}}\boldsymbol{f}$ (with entries $\partial f_i/\partial y_j$) are continuous near $(t_0, \boldsymbol{y}_0)$, then

$$\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y}), \qquad \boldsymbol{y}(t_0) = \boldsymbol{y}_0$$

has a unique local solution.

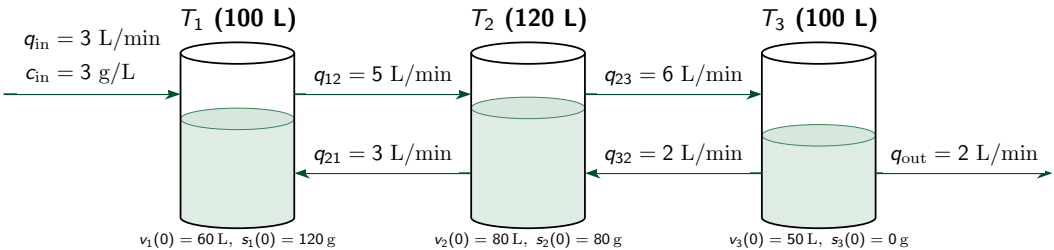
nth-order IVP

If g and its partials wrt $y, y', \ldots, y^{(n-1)}$ are continuous near the IC , then the IVP

$$y^{(n)} = g(t, y, y', \ldots, y^{(n-1)}), \qquad y^{(j)}(t_0) = a_j \quad (j = 0, \ldots, n-1)$$

has a unique local solution.

Well-Mixed Tanks



Tank Volumes

- Volume change is inflow minus outflow, in $\mathrm{L/min}$.

$$v_1' = 3 + 3 - 5 = 1 \qquad v_1(t) = 60 + t$$
$$v_2' = 5 + 2 - 3 - 6 = -2 \qquad v_2(t) = 80 - 2t$$
$$v_3' = 6 - 2 - 2 = 2 \qquad v_3(t) = 50 + 2t$$

- Volumes are in liters. The flow rates do not balance.
- Well-mixed means $c_i(t) = s_i(t)/v_i(t)$, where s_i is salt mass in grams.

When Does the Tank Model Make Sense?

- Each volume must stay positive and below capacity.

$$\begin{aligned} 0 < 60 + t < 100 &\implies 0 \leq t < 40 \\ 0 < 80 - 2t < 120 &\implies 0 \leq t < 40 \\ 0 < 50 + 2t < 100 &\implies 0 \leq t < 25 \end{aligned}$$

- Tank 3 fills first, at $t = 25$ minutes.
- The model applies for $0 \leq t < 25$. Overflow changes things!

Salt Flow Rate

- A flow carries salt at rate qc.

$$\frac{\not{\text{L}}}{\text{min}}\frac{\text{g}}{\not{\text{L}}} = \frac{\text{g}}{\text{min}}$$

  - For instance, the external flow brings salt into tank 1 at

$$q_{\text{in}}c_{\text{in}} = \left(3\,\frac{\not{\text{L}}}{\text{min}}\right)\left(3\,\frac{\text{g}}{\not{\text{L}}}\right) = 9\,\frac{\text{g}}{\text{min}}.$$

  - For instance, the pipe from tank 1 to tank 2 brings salt into tank 2 at the rate

$$q_{12}c_1(t) = \frac{5s_1(t)}{v_1(t)} = \frac{5s_1(t)}{\underbrace{60+t}_{\text{volume } v_1(t)}} \quad \text{g/min}.$$

    The same pipe sucks salt out of tank 1 at $-q_{12}c_1(t) = -5s_1(t)/(60+t)\,\text{g/min}$.

Salt Balances

- Salt mass change is salt inflow minus salt outflow.

$$s_1' = 9 + 3c_2 - 5c_1$$
$$s_2' = 5c_1 + 2c_3 - 9c_2$$
$$s_3' = 6c_2 - 4c_3$$

- Substitute the source tank's concentration:

$$c_1 = s_1/(60 + t), \qquad c_2 = s_2/(80 - 2t), \qquad c_3 = s_3/(50 + 2t).$$

- Salt rates are in $\mathrm{g/min}$. Concentrations are in $\mathrm{g/L}$.

Initial Salt Masses

- The diagram gives initial salt masses directly:

$$s_1(0) = 120 \text{ g}, \qquad s_2(0) = 80 \text{ g}, \qquad s_3(0) = 0 \text{ g}.$$

- Concentration is salt mass divided by volume. For example,

$$c_1(0) = \frac{120\,\text{g}}{60\,\text{L}} = 2\,\text{g/L}.$$

- Thus $\boldsymbol{s}_0 = (120, 80, 0)$ in grams.

Tank System as a Vector IVP

$$\boldsymbol{s}' = \boldsymbol{A}(t)\boldsymbol{s} + \boldsymbol{b}, \qquad \boldsymbol{s}(0) = \begin{pmatrix} 120 \\ 80 \\ 0 \end{pmatrix}$$

$$\boldsymbol{A}(t) = \begin{pmatrix} -5/(60+t) & 3/(80-2t) & 0 \\ 5/(60+t) & -9/(80-2t) & 2/(50+2t) \\ 0 & 6/(80-2t) & -4/(50+2t) \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 9 \\ 0 \\ 0 \end{pmatrix}.$$

- $\boldsymbol{A}(t)$ varies with time because the volumes change.
- Entries of $\boldsymbol{A}$ have units min^{-1}. Entries of $\boldsymbol{b}$ have units $\mathrm{g/min}$.
- The physical model applies for $0 \le t < 25$ minutes.

Total Salt Balance

- Add the three salt balances. Internal transfers cancel.

$$(s_1 + s_2 + s_3)' = 9 - 2c_3 = 9 - 2s_3/(50 + 2t).$$

- Total salt changes only through the external inlet and outlet.

Linear ODEs

- $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{b}(t)$ is a **Linear ODE**. Coefficients are functions of t but not $\boldsymbol{y}$.
  - It is **Constant Coefficient** (CC) if A and b do not depend on t.
  - It is **Homogeneous** if the **forcing** term $\boldsymbol{b}(t) = 0$.
- From Calculus One we know the solution of $y' = ay + 0$ is $y = y(0)e^{at}$. From Calculus Two we know the definition

$$e^{at} = 1 + (at) + \frac{1}{2!}(at)^2 + \frac{1}{2!}(at)^3 + \cdots = \sum_{k=0}^{\infty} \frac{1}{k!}(at)^k$$

- $\boldsymbol{y}(t) = e^{t\boldsymbol{A}}\boldsymbol{y}(0)$ solves the CC homogeneous system $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y} + \boldsymbol{0}$ where

$$e^{t\boldsymbol{A}} = \boldsymbol{I} + t\boldsymbol{A} + \frac{t^2\boldsymbol{A}^2}{2!} + \frac{t^3\boldsymbol{A}^3}{3!} + \cdots = \sum_{j=0}^{\infty} \frac{t^k\boldsymbol{A}^k}{k!}$$